

DEC Lab Manual & Solved Exam Compendium

Comprehensive First-Principles Lecture Notes

PART 4: Master University Examination & Numerical Problem Compendium

Purpose and Pedagogy of this Problem Compendium

In Rashtrasant Tukadoji Maharaj Nagpur University (RTMNU) and affiliated engineering university examinations for **Digital Electronics Circuits (BEL5T16C)**, the 70-mark university paper rigorously balances theoretical derivations with numerical circuit synthesis.

This section presents step-by-step, fully worked solutions to the most challenging, frequently recurring examination problems across all 5 units. Every problem is solved following the engineering standard: 1. **Given Data & Parameter Identification** 2. **First-Principles Governing Formula / Truth Table** 3. **Step-by-Step Algebraic / Circuit Derivation** 4. **Physical Sanity Check & Verification**

UNIT I: Solved Numerical & Design Problems

Problem 1.1: Signed 2's Complement Arithmetic & Hardware Overflow Detection

Question: Perform the following arithmetic operations using 8-bit 2's complement representation. In each case, determine whether an arithmetic overflow has occurred and explain the physical hardware condition used by the ALU to detect it: 1. $(+75)_{10} + (+68)_{10}$ 2. $(-42)_{10} + (-95)_{10}$ 3. $(+95)_{10} - (+42)_{10}$

Solution:

Step 1: Bit Budget and Representation

An 8-bit signed register stores numbers in the range:

$$\text{Range} = \leq \text{ft}[-2^7, +2^7 - 1\text{right}] = [-128, +127]$$

- Convert absolute magnitudes to 8-bit binary:
- $75_{10} = 64 + 8 + 2 + 1 = 01001011_2$
- $68_{10} = 64 + 4 = 01000100_2$
- $42_{10} = 32 + 8 + 2 = 00101010_2$
- $95_{10} = 64 + 16 + 8 + 4 + 2 + 1 = 01011111_2$
- Form negative numbers using 2's complement (invert all bits and add 1):
- $(-42)_{10}$: Invert $00101010 \rightarrow 11010101$, add 1 $\rightarrow \mathbf{11010110_2}$
- $(-95)_{10}$: Invert $01011111 \rightarrow 10100000$, add 1 $\rightarrow \mathbf{10100001_2}$

Part 1: $(+75)_{10} + (+68)_{10}$

Carry in:	1 0 0 0 0 0 0 0
(+75):	0 1 0 0 1 0 1 1

```
(+68):      + 0 1 0 0 0 1 0 0
-----
Sum:        1 0 0 0 1 1 1 1   (Bit 7 = 1, meaning NEGATIVE!)
```

- **Overflow Analysis:**
- Expected decimal sum: $+75 + 68 = +143_{10}$.
- But the maximum positive value an 8-bit register can hold is $+127_{10}$!
- We added two positive operands (Sign bits both 0) and received a negative result (Sign bit 1).
- **Hardware Rule Check:** In the ALU, carry into the sign bit ($C_7 = 1$) is XORed with carry out of the sign bit ($C_8 = 0$):

$$V = C_7 \oplus C_8 = 1 \oplus 0 = \mathbf{1} \quad \blacksquare \text{ OVERFLOW DETECTED!}$$

- The bit pattern 10001111 in 2's complement evaluates to $-(01110000 + 1) = -113_{10}$, which is mathematically erroneous due to register overflow.

Part 2: $(-42)_{10} + (-95)_{10}$

```
Carry in:    1 0 0 0 0 0 0 0 0
(-42):       1 1 0 1 0 1 1 0
(-95):       + 1 0 1 0 0 0 0 1
-----
Sum:         1 0 1 1 1 0 1 1 1   (Carry out C8 = 1 is discarded in 2's comp)
```

- **Overflow Analysis:**
- Expected decimal sum: $-42 + (-95) = -137_{10}$.
- The minimum negative value an 8-bit register can hold is -128_{10} !
- We added two negative numbers (Sign bits both 1) and produced a positive result ($01110111_2 = +119_{10}$)!
- **Hardware Rule Check:** Carry into sign bit $C_7 = 0$, carry out $C_8 = 1$:

$$V = C_7 \oplus C_8 = 0 \oplus 1 = \mathbf{1} \quad \blacksquare \text{ OVERFLOW DETECTED!}$$

Part 3: $(+95)_{10} - (+42)_{10} = (+95)_{10} + (-42)_{10}$

```
Carry in:    1 1 1 0 1 1 0 0 0
(+95):       0 1 0 1 1 1 1 1
(-42):       + 1 1 0 1 0 1 1 0
-----
Sum:         1 0 0 1 1 0 1 0 1   (Carry out C8 = 1 discarded)
```

- Discard end carry $C_8 = 1$. The 8-bit result is 00110101.
- Convert to decimal: $32 + 16 + 4 + 1 = +53_{10}$.
- Expected: $95 - 42 = +53_{10}$.
- **Hardware Rule Check:** $C_7 = 1$, $C_8 = 1 \quad \blacksquare \quad V = 1 \oplus 1 = \mathbf{0}$ (No overflow). Result is exact.

Problem 1.2: DC Noise Margin and Interfacing Pull-Up Resistor Calculation

Question: A standard TTL gate (7400 series) with parameters:

$$V_{OH(min)} = 2.4 \text{ V}, \quad V_{OL(max)} = 0.4 \text{ V}, \quad V_{IH(min)} = 2.0 \text{ V}, \quad V_{IL(max)} = 0.8 \text{ V}$$

$$I_{OH(max)} = -400 \text{ } \mu\text{A}, \quad I_{OL(max)} = 16 \text{ mA}, \quad I_{IH(max)} = 40 \text{ } \mu\text{A}, \quad I_{IL(max)} = -1.6 \text{ mA}$$

is used to drive a high-speed CMOS gate (74HC series powered at $V_{DD} = 5.0 \text{ V}$) having:

$$V_{IH(min)} = 3.5 \text{ V}, \quad V_{IL(max)} = 1.0 \text{ V}, \quad I_{in} \approx \pm 1 \text{ } \mu\text{A}$$

1. Show why direct interfacing fails in the HIGH state.

2. Calculate the allowable range ($R_{p(\min)}$ to $R_{p(\max)}$) for an external pull-up resistor connected from the TTL output to +5 V.

Solution:

Step 1: Why Direct Connection Fails

- Look at the LOW state:

$$V_{OL(\max)} = 0.4 \text{ V} < V_{IL(\max, \text{CMOS})} = 1.0 \text{ V}$$

$NM_L = 1.0 \text{ V} - 0.4 \text{ V} = +0.6 \text{ V}$. The LOW state communicates safely. - Look at the HIGH state:

$$V_{OH(\min, \text{TTL})} = 2.4 \text{ V} < V_{IH(\min, \text{CMOS})} = 3.5 \text{ V}$$

$$NM_H = 2.4 \text{ V} - 3.5 \text{ V} = -1.1 \text{ V} \text{ (}\mathbf{\text{NEGATIVE NOISE MARGIN!}}\text{)}$$

The CMOS gate will see 2.4 V as an indeterminate voltage in the forbidden threshold region, causing severe oscillations or intermediate conduction where both PMOS and NMOS conduct simultaneously, overheating the chip! Direct connection is impossible.

Step 2: Calculation of Maximum Pull-Up Resistance ($R_{p(\max)}$)

When the TTL output is in the HIGH state, the pull-up resistor must supply the leakage currents while maintaining the node voltage above $V_{IH(\min, \text{CMOS})} = 3.5 \text{ V}$:

$$V_{CC} - I_{\text{total leakage}} \cdot R_p \geq V_{IH(\min)}$$

The leakage current is the sum of TTL reverse leakage ($I_{OH} \approx 100 \mu\text{A}$) and CMOS input leakage ($1 \mu\text{A}$):

$$I_{\text{leakage}} \approx 100 \mu\text{A} + 1 \mu\text{A} = 101 \mu\text{A}$$

$$R_{p(\max)} = \frac{V_{CC} - V_{IH(\min)}}{I_{\text{leakage}}} = \frac{5.0 \text{ V} - 3.5 \text{ V}}{101 \times 10^{-6} \text{ A}} = \frac{1.5 \text{ V}}{101 \mu\text{A}} \approx \mathbf{14.85 \text{ k}\Omega}$$

Step 3: Calculation of Minimum Pull-Up Resistance ($R_{p(\min)}$)

When the TTL output transitions to the LOW state, its lower pull-down transistor (Q_1) turns ON and sinks current from the pull-up resistor to ground. To prevent the LOW voltage from rising above $V_{OL(\max)} = 0.4 \text{ V}$, the current sinking capacity ($I_{OL(\max)} = 16 \text{ mA}$) must not be exceeded:

$$I_{\text{sink}} = \frac{V_{CC} - V_{OL(\max)}}{R_p} + N \cdot |I_{IL}| \leq I_{OL(\max)}$$

For a single CMOS load, $|I_{IL}| \approx 0$:

$$R_{p(\min)} = \frac{V_{CC} - V_{OL(\max)}}{I_{OL(\max)}} = \frac{5.0 \text{ V} - 0.4 \text{ V}}{16 \text{ mA}} = \frac{4.6 \text{ V}}{0.016 \text{ A}} = \mathbf{287.5 \Omega}$$

Engineering Conclusion:

Any standard resistor between $1 \text{ k}\Omega$ and $10 \text{ k}\Omega$ (typically $2.2 \text{ k}\Omega$ or $3.3 \text{ k}\Omega$) guarantees safe, fast, and robust TTL-to-CMOS interfacing.

UNIT II: Solved Boolean Minimization & K-Map Problems

Problem 2.1: 4-Variable K-Map Minimization with Don't Cares

Question: Minimize the following 4-variable Boolean logic function in both Sum-of-Products (SOP) and Product-of-Sums (POS) standard forms. Explicitly identify all Prime Implicants (PI) and Essential Prime Implicants (EPI):

$$f(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + \sum d(0, 2, 5)$$

Solution:

Step 1: Populate the 4-Variable Karnaugh Map

The map coordinates follow Gray code sequence (00, 01, 11, 10):

	CD			
AB	00	01	11	10
00	+-----+-----+-----+			
	X0 11 13 X2			
	+-----+-----+-----+			
01	4 X5 17 6			
	+-----+-----+-----+			
11	12 13 115 14			
	+-----+-----+-----+			
10	8 9 111 10			
	+-----+-----+-----+			

- Cell 1, 3, 7, 11, 15 contain 1.
- Cell 0, 2, 5 contain x (Don't Care).
- All remaining cells (4, 6, 8, 9, 10, 12, 13, 14) contain 0.

Step 2: Form Optimal SOP Groupings (Powers of 2: 8, 4, 2)

1. **Octet Exploration:** Look at the four corners and edges. Can we form an 8-group? No.
2. **Quad 1 (The Entire Column CD = 11):**
Cells (3, 7, 15, 11) all contain 1!
This vertical column spans all four rows (AB = 00, 01, 11, 10).
Variables A and B eliminate completely.
Term 1 = C D
3. **Quad 2 (Row 00):**
Look at row AB = 00: cells are (0, 1, 3, 2) having values (X, 1, 1, X).
By treating the don't cares at cells 0 and 2 as 1s, we form a 4-cell group!
In this row, A=0 and B=0. Variables C and D span all combinations and eliminate.
Term 2 = A B
4. **Coverage Check:**
5. Cell 1 is covered by Quad 2.
6. Cell 3 is covered by both Quad 1 and Quad 2.
7. Cells 7, 11, 15 are covered by Quad 1.
8. All specified minterms (1, 3, 7, 11, 15) are 100% covered!
9. What about cell 5 (X)? We leave it as 0 because grouping it would require an extra redundant gate.

Prime Implicant Inventory:

- **Prime Implicants (PI):** C D, A B, A D (cells 1, 3, 5, 7).
- **Essential Prime Implicants (EPI):**
- C D is essential because minterms 7, 11, 15 are covered by no other prime implicant.
- A B is essential because minterm 1 is covered only by it (if we choose the minimal 2-term cover).
- **Minimal SOP Equation:**

$$\mathbf{f(A, B, C, D) = A B + C D}$$

Step 3: Minimal POS Minimization (Grouping the 0s)

The 0 cells are: **4, 6, 8, 9, 10, 12, 13, 14**. We group the zeros together with don't cares (X): 1. **Quad of 0s at left column (CD = 00)**: Cells (**4, 12**) with (**0, X**)? No, cell 0 is X.
 Let's group cells (**4, 6, 12, 14**): In these cells, **B=1** and **D=0**.
 Complement: (**B + D**). 2. **Quad of 0s at row 10 and 11 (AB = 10, 11 ■ A=1)**:
 Cells (**8, 9, 12, 13**): **A=1** and **C=0**.
 Complement: (**A + C**). 3. **Minimal POS Equation**:

$$\mathbf{f(A, B, C, D) = (A + C)(B + D)}$$

(Both forms require only two 2-input gates plus an output gate, representing an 85% hardware reduction from the unsimplified 5-AND SOP expression!)

UNIT III: Solved Combinational Logic Synthesis Problems

Problem 3.1: Full Adder Implementation Using Only 4-to-1 Multiplexers

Question: Design and realize a Full Adder (Outputs: Sum **S** and Carry **C_{out}**) using **only two 4-to-1 Multiplexers** (such as the dual 74153 IC) and no external logic gates. Inputs are **A, B, C_{in}**.

Solution:

Step 1: Full Adder Truth Table

A	B	C _{in}	Sum (S)	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Step 2: Multiplexer Input Assignment

A 4-to-1 MUX has two select inputs (**S₁, S₀**) and four data inputs (**I₀, I₁, I₂, I₃**). - We assign variables **A** and **B** to the select lines:

$$\mathbf{S_1 = A, S_0 = B}$$

- The remaining input variable **C_{in}** (and its complement $\overline{\mathbf{C_{in}}}$, 0, or 1) will be connected directly to the data inputs **I₀, I₁, I₂, I₃**.

Step 3: Deriving Data Inputs for MUX 1 (Sum S)

- When **A B = 00**:
- If **C_{in} = 0 ■ S = 0**
- If **C_{in} = 1 ■ S = 1**
- Therefore: $\mathbf{I_0 = C_{in}}$

- When $A B = 01$:
- If $C_{in} = 0 \blacksquare S = 1$
- If $C_{in} = 1 \blacksquare S = 0$
- Therefore: $\mathbf{I_1 = \overline{C_{in}}}$
- When $A B = 10$:
- If $C_{in} = 0 \blacksquare S = 1$
- If $C_{in} = 1 \blacksquare S = 0$
- Therefore: $\mathbf{I_2 = \overline{C_{in}}}$
- When $A B = 11$:
- If $C_{in} = 0 \blacksquare S = 0$
- If $C_{in} = 1 \blacksquare S = 1$
- Therefore: $\mathbf{I_3 = C_{in}}$

Step 4: Deriving Data Inputs for MUX 2 (Carry C_{out})

- When $A B = 00$:
- For both $C_{in} = 0$ and $C_{in} = 1$, $C_{out} = 0$.
- Therefore: $\mathbf{I_0 = 0}$ (tied to Ground)
- When $A B = 01$:
- If $C_{in} = 0 \blacksquare C_{out} = 0$
- If $C_{in} = 1 \blacksquare C_{out} = 1$
- Therefore: $\mathbf{I_1 = C_{in}}$
- When $A B = 10$:
- If $C_{in} = 0 \blacksquare C_{out} = 0$
- If $C_{in} = 1 \blacksquare C_{out} = 1$
- Therefore: $\mathbf{I_2 = C_{in}}$
- When $A B = 11$:
- For both $C_{in} = 0$ and $C_{in} = 1$, $C_{out} = 1$.
- Therefore: $\mathbf{I_3 = 1}$ (tied to V_{CC})

Synthesis Summary:

Both functions are realized using a single dual 4-to-1 MUX IC (74153) with zero external gates: - **MUX 1 (Sum):** $S_1 = A$, $S_0 = B$, Inputs: ($I_0 = C_{in}$, $I_1 = \overline{C_{in}}$, $I_2 = \overline{C_{in}}$, $I_3 = C_{in}$). - **MUX 2 (Carry):** $S_1 = A$, $S_0 = B$, Inputs: ($I_0 = 0$, $I_1 = C_{in}$, $I_2 = C_{in}$, $I_3 = 1$).

UNIT IV: Solved Sequential Circuit Conversion & Counter Design Problems

Problem 4.1: Systematic Conversion of an SR Flip-Flop to a JK Flip-Flop

Question: Design the combinational conversion logic required to convert an available **SR Flip-Flop** into a functional **JK Flip-Flop**.

Solution:

Step 1: Master Conversion Table

We list the desired inputs (**J**, **K**), present state (Q_n), next state (Q_{n+1}), and the required SR excitation values from the SR excitation table:

J	K	Q_n	Desired Q_{n+1}	Required S	Required R
0	0	0	0	0	x
0	0	1	1	x	0

J	K	Q _n	Desired Q _{n+1}	Required S	Required R
0	1	0	0	0	x
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	x	0
1	1	0	1	1	0
1	1	1	0	0	1

Step 2: K-Map Minimization for S Input

Inputs to K-map are **J**, **K**, **Q_n**:

	KQn			
J	00	01	11	10
0	+-----+			
	0	X	0	0
	+-----+			
1	1	X	0	1
	+-----+			

- Grouping cells **(4, 6)** [where **J=1** and **Q_n=0**]:

$$\mathbf{S = J \, Q_n}$$

Step 3: K-Map Minimization for R Input

	KQn			
J	00	01	11	10
0	+-----+			
	X	0	1	X
	+-----+			
1	0	0	1	0
	+-----+			

- Grouping cells **(3, 7)** [where **K=1** and **Q_n=1**]:

$$\mathbf{R = K \, Q_n}$$

Hardware Reality Check:

Feed input **J** into a 2-input AND gate with the complement output **Q**, and connect its output to pin **S**. Feed input **K** into a 2-input AND gate with output **Q**, and connect its output to pin **R**.

Notice what this did physically: When **J=K=1** and **Q=1**, **R** receives **1** and **S** receives **0**, forcing the flip-flop to RESET to **0** on the clock edge. When **Q=0**, **S** receives **1** and **R** receives **0**, forcing it to SET to **1**. The forbidden condition (**S=R=1**) is physically impossible because **Q** and **Q** can never be **1** simultaneously!

UNIT V: Solved Converter & Memory Architecture Problems

Problem 5.1: 8-Bit R-2R Ladder DAC Performance & Resolution

Question: An 8-bit R-2R ladder Digital-to-Analog Converter is powered with an analog reference voltage $V_{\text{ref}} = +10.0 \text{ V}$.
1. Calculate the resolution (step size ΔV) in millivolts. 2. Determine the analog output voltage for a digital input code of **10100110**. 3. Calculate the full-scale output voltage (V_{FS}).

Solution:

Step 1: Resolution (Step Size)

An 8-bit DAC has $n=8$ bits, giving $2^8 - 1 = 255$ discrete conversion intervals:

$$\Delta V = \frac{V_{\text{ref}}}{2^n - 1} = \frac{10.0 \text{ V}}{255} = 0.039215 \text{ V} = \mathbf{39.22 \text{ mV per step}}$$

(Note: In binary weighted converters referenced to an unscaled op-amp buffer, step size is often formulated as $V_{\text{ref}}/2^n = 10.0/256 = 39.06 \text{ mV}$. Both conventions are accepted in university evaluation when explicitly stated).

Step 2: Output Voltage for Digital Code 10100110

Convert binary code to decimal:

$$D = (1 \times 128) + (0 \times 64) + (1 \times 32) + (0 \times 16) + (0 \times 8) + (1 \times 4) + (1 \times 2) + (0 \times 1)$$

$$D = 128 + 32 + 4 + 2 = 166_{10}$$

The analog output voltage is:

$$V_{\text{out}} = D \times \Delta V = 166 \times 39.215 \text{ mV} = \mathbf{6.510 \text{ V}}$$

Alternatively, evaluate via node current summation:

$$V_{\text{out}} = V_{\text{ref}} \leq \text{ft}((1 / 2) + (0 / 4) + (1 / 8) + (0 / 16) + (0 / 32) + (1 / 64) + (1 / 128) + (0 / 256)\text{right})$$

$$V_{\text{out}} = 10.0 \leq \text{ft}(0.5 + 0.125 + 0.015625 + 0.0078125\text{right}) = 10.0 \times 0.6484375 = \mathbf{6.484 \text{ V}}$$

Step 3: Full-Scale Output Voltage (V_{FS})

When all input bits are **1** (**11111111** = **255**₁₀):

$$V_{\text{FS}} = (255 / 256) \times 10.0 \text{ V} = \mathbf{9.961 \text{ V}}$$

Notice that the full-scale analog output never reaches **10.0 V** exactly—it is always exactly **1 LSB less than V_{ref}** !

Problem 5.2: Memory System Expansion (**4K × 8** using **1K × 4** Chips)

Question: Design a complete memory module with a capacity of **4K × 8 bits** using individual **1K × 4** static RAM chips. Show the address decoding circuit using a 2-to-4 line active-low decoder.

Solution:

Step 1: Calculate Required Chips

- Total memory capacity needed: **4K × 8 = 4096 × 8 = 32,768 bits**.
- Capacity per chip: **1K × 4 = 1024 × 4 = 4,096 bits**.

$$\mathbf{\text{Total Chips Required} = \frac{4\text{K} \times 8}{1\text{K} \times 4} = 4 \times 2 = \mathbf{8 \text{ chips}}}$$

Step 2: Address and Data Bus Architecture

- **Total Address Lines Needed:**

$$4K = 4096 = 2^{12} \blacksquare \text{mathbf{12 Address Lines (A}_{11} \text{ to A}_0\text{)}}$$

- **Total Data Lines Needed:**

$$8 \text{ bits } \blacksquare \text{mathbf{8 Data Lines (D}_7 \text{ to D}_0\text{)}}$$

- **Address Lines for Individual Chips:**

$$1K = 1024 = 2^{10} \blacksquare \text{mathbf{10 Address Lines (A}_9 \text{ to A}_0\text{)}}$$

The lower 10 address lines (**A₀** through **A₉**) connect directly to all 8 memory chips in parallel. - **Address Lines for Chip Selection:** The upper 2 address lines (**A₁₁**, **A₁₀**) determine which pair of chips is activated:

$$2^2 = 4 \text{ banks of memory}$$

Step 3: Chip Bank Arrangement

We arrange the 8 chips into **4 banks**, with each bank containing **2 chips** wired in parallel to provide the full 8-bit word width: - **Bank 0 (Addresses 000H to 3FFH):** - Chip 1 provides Data bits **D₀ - D₃**. - Chip 2 provides Data bits **D₄ - D₇**. - Enabled by Decoder Output **Y₀ (A₁₁A₁₀ = 00)**. - **Bank 1 (Addresses 400H to 7FFH):** - Chip 3 provides Data bits **D₀ - D₃**. - Chip 4 provides Data bits **D₄ - D₇**. - Enabled by Decoder Output **Y₁ (A₁₁A₁₀ = 01)**. - **Bank 2 (Addresses 800H to BFFH):** - Chip 5 provides Data bits **D₀ - D₃**. - Chip 6 provides Data bits **D₄ - D₇**. - Enabled by Decoder Output **Y₂ (A₁₁A₁₀ = 10)**. - **Bank 3 (Addresses C00H to FFFH):** - Chip 7 provides Data bits **D₀ - D₃**. - Chip 8 provides Data bits **D₄ - D₇**. - Enabled by Decoder Output **Y₃ (A₁₁A₁₀ = 11)**.

Step 4: Address Decoder Connection

A 2-to-4 line active-low decoder (such as half of a 74LS139) is connected as follows: - Decoder Select Input **A = A₁₀** - Decoder Select Input **B = A₁₁** - Decoder Enable **G** tied to Ground (permanently enabled). - Decoder output **Y₀** connects to Chip Select **CS** of Chips 1 and 2. - Decoder output **Y₁** connects to Chip Select **CS** of Chips 3 and 4. - Decoder output **Y₂** connects to Chip Select **CS** of Chips 5 and 6. - Decoder output **Y₃** connects to Chip Select **CS** of Chips 7 and 8. - The master Write Enable line (**WE**) connects to pin **WE** of all 8 chips simultaneously.

*This concludes the complete hardware architecture for the **4K × 8** memory system.*

PART 5: Practical Digital Electronics Laboratory Manual & Viva-Voce Compendium

Purpose & Structure of the Laboratory Manual

The syllabus for **BEL5T16C** allocates 1 hour per week of Practical / Activity (T/A). Laboratory verification bridges theoretical logic equations and physical silicon chips mounted on a breadboard or trainer kit.

This section provides complete, standardized laboratory experiment worksheets for 10 core digital experiments. Each experiment is structured with: 1. **Aim of the Experiment** 2. **Components & Equipment Required** (with exact TTL IC part numbers) 3. **Pin Configuration & Breadboard Wiring Guide** 4. **Step-by-Step Laboratory Procedure** 5. **Truth / Observation Table** 6. **Viva-Voce Questions with Detailed Technical Answers** (frequently asked in practical oral exams)

EXPERIMENT 1: Study and Verification of Truth Tables of Basic and Universal Logic Gates

1.1 Aim

To experimentally verify the truth tables of basic logic gates (AND, OR, NOT) and universal logic gates (NAND, NOR) using 7400-series TTL integrated circuits.

1.2 Components Required

- IC 7404 (Hex Inverter)
- IC 7408 (Quad 2-input AND Gate)
- IC 7432 (Quad 2-input OR Gate)
- IC 7400 (Quad 2-input NAND Gate)
- IC 7402 (Quad 2-input NOR Gate)
- Digital Logic Trainer Kit with +5 V regulated DC power supply
- Logic input toggle switches (0 V and +5 V)
- Logic output indicator LEDs with internal current-limiting resistors
- Breadboard and single-strand jumper wires

1.3 IC Pin Configuration & Hardware Grounding

- **Power Connections:** For all 14-pin 7400-series ICs (except 7402):
- **Pin 14:** Connect to +5 V DC (V_{CC}).
- **Pin 7:** Connect to Ground (0 V).
- *Warning:* Never power a TTL IC with more than 5.25 V; reverse polarity will instantly destroy the internal bipolar junctions.
- **Pin Mapping:**
- **7404:** Pin 1 = A, Pin 2 = A; Pin 3 = B, Pin 4 = B; etc.
- **7408:** Pin 1 = A, Pin 2 = B, Pin 3 = Y = A · B; Pin 4 = C, Pin 5 = D, Pin 6 = Y = C · D.
- **7432:** Pin 1 = A, Pin 2 = B, Pin 3 = Y = A + B.
- **7400:** Pin 1 = A, Pin 2 = B, Pin 3 = Y = A · B.
- **7402 (Special Pinout):** Pin 1 = Output Y = A + B, Pin 2 = Input A, Pin 3 = Input B!

1.4 Step-by-Step Procedure

1. Mount the IC gently onto the breadboard straddling the center divider.
2. Connect Pin 14 to the +5 V rail and Pin 7 to the Ground rail.
3. Connect the input pins to digital toggle switches S₁ and S₂.
4. Connect the gate output pin to an LED logic monitor.
5. Apply all input combinations (00, 01, 10, 11) and record the state of the LED (1 = ON/Lit, 0 = OFF/Extinguished).

1.5 Observation Table

Gate	Type	Input A	Input B	Expected Output	Measured LED State
NOT (7404)	Inverter	0	—	1	ON
		1	—	0	OFF
AND (7408)	Product	0	0	0	OFF
		0	1	0	OFF
		1	0	0	OFF
		1	1	1	ON
OR (7432)	Sum	0	0	0	OFF
		0	1	1	ON
		1	0	1	ON

Gate	Type	Input A	Input B	Expected Output	Measured LED State
		1	1	1	ON
NAND (7400)	Universal	0	0	1	ON
		0	1	1	ON
		1	0	1	ON
		1	1	0	OFF
NOR (7402)	Universal	0	0	1	ON
		0	1	0	OFF
		1	0	0	OFF
		1	1	0	OFF

1.6 Viva-Voce Questions & Technical Answers

- Q1: Why are NAND and NOR gates called "universal gates"?**
Answer: Because any arbitrary Boolean logic function—including the basic operations AND, OR, and NOT—can be realized using only NAND gates or only NOR gates without requiring any other gate type.
- Q2: What happens if an unused input of a standard TTL NAND gate is left unconnected (floating)?**
Answer: In standard bipolar TTL, an unconnected input floats to an internal logic HIGH level ($\approx 1.4\text{ V}$ to 1.8 V) because no current is drawn out of the multi-emitter base junction. However, leaving pins floating makes the circuit highly vulnerable to capacitive electromagnetic noise pickup. Good engineering practice mandates tying unused inputs to V_{CC} through a $1\text{ k}\Omega$ pull-up resistor or tying them to an active input in parallel.
- Q3: What is the propagation delay of a standard 7400 TTL gate?**
Answer: Typically around 10 ns (nanoseconds). In Schottky TTL (74S) it drops to 3 ns , and in Low-Power Schottky (74LS) it is approximately 9.5 ns .

EXPERIMENT 2: Design and Hardware Implementation of Half Adder and Full Adder

2.1 Aim

To design, assemble, and test a Half Adder and a Full Adder circuit using basic logic gates (XOR, AND, OR).

2.2 Components Required

- IC 7486 (Quad 2-input XOR Gate)
- IC 7408 (Quad 2-input AND Gate)
- IC 7432 (Quad 2-input OR Gate)
- Logic Trainer Kit, Connecting Wires, Multimeter

2.3 Circuit Design & Governing Equations

1. Half Adder Equations:

Sum $S = A \oplus B$

Carry $C = A \cdot B$

- Hardware: One gate of 7486 (pins 1, 2 $\rightarrow$ 3) + one gate of 7408 (pins 1, 2 $\rightarrow$ 3).
- Full Adder Equations:**

$$\text{Sum } S = A \oplus B \oplus C_{\text{in}}$$

$$\text{Carry } C_{\text{out}} = A \cdot B + C_{\text{in}} \cdot (A \oplus B)$$

- Hardware: Two XOR gates (7486), two AND gates (7408), and one OR gate (7432).

2.4 Wiring and Testing Procedure

- Wire the Half Adder circuit on the breadboard. Connect inputs **A** and **B** to data switches.
- Verify that the Sum LED illuminates when either **A** or **B** is HIGH, but turns OFF when both are HIGH while the Carry LED turns ON.
- Expand to the Full Adder circuit by cascading the intermediate sum $(A \oplus B)$ into the second XOR gate along with carry input C_{in} .
- Feed $(A \cdot B)$ and $C_{\text{in}} \cdot (A \oplus B)$ into the OR gate to generate C_{out} .
- Record all 8 combinations of **A**, **B**, C_{in} .

2.5 Full Adder Observation Table

A	B	C_{in}	Expected S	Expected C_{out}	Measured S (LED)	Measured C_{out} (LED)
0	0	0	0	0	OFF	OFF
0	0	1	1	0	ON	OFF
0	1	0	1	0	ON	OFF
0	1	1	0	1	OFF	ON
1	0	0	1	0	ON	OFF
1	0	1	0	1	OFF	ON
1	1	0	0	1	OFF	ON
1	1	1	1	1	ON	ON

2.6 Viva-Voce Questions & Technical Answers

- Q1: Can a Full Adder be constructed using two Half Adders and an OR gate?**
Answer: Yes. Half Adder 1 computes partial sum $S_1 = A \oplus B$ and carry $C_1 = A \cdot B$. Half Adder 2 adds S_1 and C_{in} to produce final sum $S = S_1 \oplus C_{\text{in}}$ and carry $C_2 = S_1 \cdot C_{\text{in}}$. The final carry is $C_{\text{out}} = C_1 + C_2$.
- Q2: Why is the carry expression simplified with an OR gate rather than an XOR gate?**
Answer: Because $C_1 (A \cdot B)$ and $C_2 [(A \oplus B) \cdot C_{\text{in}}]$ are mutually exclusive—they can never be 1 at the same physical instant! When $A=B=1$, $A \oplus B = 0$, forcing $C_2 = 0$. Therefore, $C_1 + C_2$ is strictly identical to $C_1 \oplus C_2$.

EXPERIMENT 3: 4-Bit Binary Parallel Adder / Subtractor Using IC 7483

3.1 Aim

To implement and verify a 4-bit parallel binary adder and subtractor circuit using the 7483 4-bit binary adder IC and controlled XOR inverters (IC 7486).

3.2 Components Required

- IC 7483 (4-Bit Binary Full Adder with Fast Carry)

- IC 7486 (Quad 2-input XOR Gate)
- Logic Trainer Kit, LEDs, Multimeter

3.3 Hardware Pinout of IC 7483

- **Pin 5: V_{CC} (+5 V), Pin 12: Ground (0 V).**
- **First 4-bit Operand (A):** A_1 (Pin 1), A_2 (Pin 3), A_3 (Pin 8), A_4 (Pin 11).
- **Second 4-bit Operand (B):** B_1 (Pin 16), B_2 (Pin 4), B_3 (Pin 7), B_4 (Pin 10).
- **Carry In (C_0):** Pin 13.
- **Sum Outputs:** Σ_1 (Pin 2), Σ_2 (Pin 15), Σ_3 (Pin 6), Σ_4 (Pin 9).
- **Carry Out (C_4):** Pin 14.

3.4 Circuit Operation

- Connect a **Mode Control switch (M)** to Carry In (Pin 13) and to one input of each of the four XOR gates in IC 7486.
- Connect operand bits B_1 , B_2 , B_3 , B_4 to the second input of each XOR gate.
- Connect the XOR outputs to the **B** inputs of IC 7483.
- **Addition Mode ($M = 0$):**
- $B_i \oplus 0 = B_i$ (true data passes through).
- $C_0 = 0$.
- IC 7483 performs standard binary addition: $\mathbf{A + B}$.
- **Subtraction Mode ($M = 1$):**
- $B_i \oplus 1 = \overline{B_i}$ (operand **B** is bitwise inverted).
- $C_0 = 1$ (the +1 required for 2's complement is fed into Carry In).
- IC 7483 performs 2's complement subtraction: $\mathbf{A + B + 1 = A - B}$.

3.5 Viva-Voce Questions & Technical Answers

- **Q1: In subtraction mode ($M=1$), how do you interpret the carry output C_4 ?**
Answer: In 2's complement subtraction, if $C_4 = 1$, the result is positive ($A \geq B$) and the answer appears directly in true binary form. If $C_4 = 0$, the result is negative ($A < B$) and the answer appears in 2's complement form; taking the 2's complement of the output reveals its true negative magnitude.
- **Q2: What is the advantage of using IC 7483 over cascading four individual 1-bit full adders?**
Answer: IC 7483 incorporates internal high-speed lookahead carry logic across all 4 bits, reducing the carry propagation delay from $\approx 40 \text{ ns}$ down to under 10 ns .

EXPERIMENT 4: Multiplexer & Demultiplexer Implementation (IC 74153 & IC 74138)

4.1 Aim

To study the operation of a 4-to-1 multiplexer (IC 74153) and a 3-to-8 line decoder/demultiplexer (IC 74138).

4.2 Components Required

- IC 74153 (Dual 4-to-1 Data Selector/Multiplexer)
- IC 74138 (3-to-8 Line Decoder / Demultiplexer)
- Trainer Kit, Wires, LEDs

4.3 IC 74153 Multiplexer Verification

- Connect select inputs S_1 and S_0 to toggle switches.
- Apply distinct binary signals (e.g., $I_0 = 0$, $I_1 = 1$, $I_2 = 0$, $I_3 = 1$) to data inputs.
- Tie active-low strobe $\overline{1G}$ (Pin 1) to Ground to enable the multiplexer.
- Observe output $1Y$ (Pin 7) as select lines sequence through 00 , 01 , 10 , 11 .

4.4 IC 74138 Demultiplexer Mode

- A decoder operates as a **demultiplexer** by using its data select lines (**A**, **B**, **C**) as address routing lines and using an active-low enable pin ($\overline{G_1}$ or $\overline{G_2}$) as the single serial **Data Input (D_{in})**!
- Tie enable pin G_1 HIGH (+5 V) and $\overline{G_2}$ LOW (0 V).

- Apply the data bit to `overline{line{6_{2B}}}`. The data bit appears at the selected output line (`Y_0` through `Y_7`) designated by the 3-bit binary address on `C`, `B`, `A`.

4.5 Viva-Voce Questions & Technical Answers

- **Q1: Why are the outputs of IC 74138 active-low?**
Answer: In digital systems, memory chip enable (`CE`) and peripheral select lines are standard active-low to reduce static power consumption and provide higher noise immunity against positive-going line transients.
- **Q2: Can any n -variable Boolean function be implemented using a multiplexer with $n-1$ select lines?**
Answer: Yes. By assigning $n-1$ variables to the select lines, the remaining variable (in true form, inverted form, 0, or 1) is applied directly to the data inputs.

EXPERIMENT 5: BCD to 7-Segment Display Decoder/Driver (IC 7447 & Common Anode Display)

5.1 Aim

To interface a BCD-to-7-segment decoder/driver (IC 7447) with a common anode 7-segment LED display and display decimal digits 0 through 9.

5.2 Components Required

- IC 7447 (BCD to 7-Segment Decoder/Driver with open-collector outputs)
- Common Anode 7-Segment LED Display (e.g., LTS-542A)
- Seven **330 Ω** current-limiting resistors (`rac{1}{4} W`)
- Trainer Kit, Wires

5.3 Hardware Wiring Guidelines

- **Common Anode Terminal:** Connect the common anode pins (Pins 3 and 8 of the display) to **+5 V**.
- **Current-Limiting Resistors:** Connect a separate **330 Ω** resistor in series with each segment cathode pin (`a`, `b`, `c`, `d`, `e`, `f`, `g`) between the display and the corresponding output pin of IC 7447 (pins 13 through 9).
- *Critical Warning:* Never connect a common anode LED display directly to IC 7447 outputs without series resistors! Doing so causes excessive forward current (**>50 mA**), permanently burning out the display segments.
- **Resistor Calculation:**

$$R = \frac{V_{CC} - V_F(LED) - V_{OL}(IC)}{I_F} = \frac{5.0 \text{ V} - 1.8 \text{ V} - 0.4 \text{ V}}{10 \text{ mA}} = \frac{2.8 \text{ V}}{0.010 \text{ A}} = 280 \text{ } \Omega \blacksquare \text{ Use } 330 \text{ } \Omega$$

- **Control Pins:** Tie Lamp Test (`LT`, Pin 3), Blanking Input (`BI/RB0`, Pin 4), and Ripple Blanking Input (`RBI`, Pin 5) HIGH (**+5 V**) for normal numeric decoding.

5.4 Viva-Voce Questions & Technical Answers

- **Q1: Why does IC 7447 require open-collector outputs?**
Answer: Open-collector outputs can sink significant current (up to **40 mA**) to ground when turned ON, and can tolerate pull-up voltages higher than **5 V** (up to **15 V**), allowing it to directly drive high-voltage displays and relays.
- **Q2: What is the purpose of the Lamp Test (`LT`) pin?**
Answer: When `LT` is pulsed LOW, all seven segments are forced ON simultaneously regardless of the BCD input. This allows a technician to immediately verify if any segment LED has burned out.

EXPERIMENT 6: Verification of JK and D Flip-Flops (IC 7476 & IC 7474)

6.1 Aim

To verify the truth tables, excitation behavior, and edge-triggered clocking of Master-Slave JK Flip-Flop (IC 7476) and Positive-Edge-Triggered D Flip-Flop (IC 7474).

6.2 Components Required

- IC 7476 (Dual JK Master-Slave Flip-Flop with Preset and Clear)
- IC 7474 (Dual D Positive-Edge-Triggered Flip-Flop)
- Single-pulse debounced clock generator switch
- LEDs, Trainer Kit

6.3 IC 7476 Pinout & Operation

- **Preset (PRE, Pin 2) & Clear (CLR, Pin 3):** Active-low asynchronous overrides. For normal clocked operation, connect both to **+5 V**.
- Apply logic levels to **J** (Pin 4) and **K** (Pin 16).
- Apply a single manually debounced clock pulse to **CLK** (Pin 1).
- Observe outputs **Q** (Pin 15) and **Q** (Pin 14) on falling clock transitions.

6.4 Observation Table: JK Flip-Flop

PRE	CLR	Clock	J	K	Next State Q_{n+1}	Operation
0	1	x	x	x	1	Asynchronous Preset
1	0	x	x	x	0	Asynchronous Clear
1	1	downarrow	0	0	Q_n	No Change (Hold)
1	1	downarrow	0	1	0	Reset
1	1	downarrow	1	0	1	Set
1	1	downarrow	1	1	Q_n	Toggle

6.5 Viva-Voce Questions & Technical Answers

- **Q1: Why are Preset and Clear called "asynchronous" inputs?**
Answer: Because they override the clock and data inputs immediately, forcing the flip-flop to SET or RESET instantaneously without waiting for a clock pulse transition.
- **Q2: Why must a mechanical switch be debounced when driving a flip-flop clock pin?**
Answer: Mechanical switch contacts bounce microscopically for **5 to 20 ms** upon closure, generating dozens of rapid voltage spikes. Because a flip-flop can switch in nanoseconds, it will count every bounce as a distinct clock pulse, causing erratic state transitions. Debouncing with an SR latch or Schmitt trigger eliminates this.

EXPERIMENT 7: Design of Mod-10 (Decade) Asynchronous Counter Using IC 7490

7.1 Aim

To construct and verify a Mod-10 (Decade / BCD) asynchronous ripple counter using the 7490 decade counter IC and display counts from **0** to **9** on LEDs.

7.2 Components Required

- IC 7490 (Decade Counter)
- Logic Trainer Kit with variable pulse generator (**1 Hz**)
- LEDs, Wires

7.3 Wiring Guide for BCD Count Sequence (**0 → 9**)

- **Power:** Pin 5 = +5 V, Pin 10 = Ground.
- **Reset Pins:** Connect **R_1** (Pin 2) and **R_2** (Pin 3) to Ground. Connect **S_1** (Pin 6) and **S_2** (Pin 7) to Ground.
- **Internal Structure:** IC 7490 contains a Mod-2 section (Input **A**, Pin 14; Output **Q_A**, Pin 12) and a Mod-5 section (Input **B**, Pin 1; Outputs **Q_B**, **Q_C**, **Q_D** at pins 9, 8, 11).
- **Decade Counter Wiring:** Connect output **Q_A** (Pin 12) directly to Input **B** (Pin 1)!
- Connect master clock pulses (**1 Hz**) to Input **A** (Pin 14).
- Connect outputs **Q_D**, **Q_C**, **Q_B**, **Q_A** to four LEDs (**Q_D** is MSB, **Q_A** is LSB).

7.4 Viva-Voce Questions & Technical Answers

- **Q1: How does IC 7490 automatically reset after state 9 (1001)?**

Answer: The internal Mod-5 counter transitions back to state 0 on the 5th pulse entering Input **B**. Because the Mod-2 counter divides the incoming clock by 2 before driving the Mod-5 stage, the combined circuit automatically cycles through exactly $2 \times 5 = 10$ unique states (0000 to 1001) and resets to 0000 on the 10th clock pulse.

- **Q2: How can an IC 7490 be configured as a Mod-6 counter?**

Answer: To reset at count 6 (0110), feed outputs **Q_C** (weight 4) and **Q_B** (weight 2) directly into the reset pins **R_1** and **R_2**. The instant the counter reaches 0110, pins **R_1** and **R_2** both become HIGH, triggering an immediate asynchronous clear back to 0000.

EXPERIMENT 8: 4-Bit Shift Register Implementation Using IC 7495

8.1 Aim

To configure the 7495 4-bit universal shift register in Serial-In Serial-Out (SISO) and Serial-In Parallel-Out (SIPO) modes.

8.2 Components Required

- IC 7495 (4-Bit Parallel-Access Shift Register)
- Trainer Kit, Wires, LEDs

8.3 Operation Modes

- **Mode Control Pin (Pin 6):**
- Connect Mode Control LOW (**0 V**) for Shift-Right mode.
- Connect Clock 1 (Pin 8) to the shift clock source.
- **SISO Mode:**
- Apply serial data bits to Serial Input (Pin 1).
- Clock 4 times to shift bits through stages **Q_A**, **Q_B**, **Q_C**, **Q_D**.
- Read serial output from **Q_D** (Pin 10).
- **SIPO Mode:**
- Connect four LEDs to outputs **Q_A**, **Q_B**, **Q_C**, **Q_D** (Pins 13, 12, 11, 10).
- As clock pulses arrive, observe data bits physically marching across the LED row from left to right.

8.4 Viva-Voce Questions & Technical Answers

- **Q1: How many clock pulses are required to enter and retrieve an 8-bit word in a SISO register?**

Answer: It requires 8 clock pulses to load the data serially into the register and another 8 clock pulses to shift it out serially, requiring a total of 16 clock pulses. In contrast, a PIPO register transfers the word in a single clock cycle!

EXPERIMENT 9: 4-Bit R-2R Ladder Digital-to-Analog Converter

9.1 Aim

To construct a 4-bit R-2R ladder Digital-to-Analog Converter using discrete resistors and an operational amplifier (LM741), and verify linearity and step size.

9.2 Components Required

- Op-Amp IC LM741 (8-pin DIP)

- Dual DC power supply ($\pm 12\text{ V}$ for op-amp rails)
- Six $10\text{ k}\Omega$ resistors (1% precision)
- Six $20\text{ k}\Omega$ resistors (or pairs of two $10\text{ k}\Omega$ in series)
- Digital Multimeter (DMM) with 1 mV DC resolution
- Logic toggle switches (0 V / $+5\text{ V}$)

9.3 Circuit Schematic & Connections

- Assemble the R-2R ladder on the breadboard. Connect ladder node branches to data switches **D_3** (MSB), **D_2**, **D_1**, **D_0** (LSB).
- Connect the ladder output node to the inverting input (Pin 2) of LM741.
- Connect non-inverting input (Pin 3) of LM741 to Ground (0 V).
- Connect feedback resistor **R_f** = $20\text{ k}\Omega$ between output (Pin 6) and inverting input (Pin 2).
- Connect Pin 7 to $+12\text{ V}$ and Pin 4 to -12 V .

9.4 Observation & Linearity Table (V_{ref} = 5.0 V)

Theoretical Step Size $\Delta V = \frac{5.0\text{ V}}{16} = 0.3125\text{ V} = 312.5\text{ mV}$				
Binary Input (D_3 D_2 D_1 D_0)	Decimal Equivalent	Theoretical V_{out} (V)	Measured V_{out} (DMM)	Absolute Error (mV)
0000	0	0.000	0.002	2
0001	1	-0.313	-0.315	2
0010	2	-0.625	-0.628	3
0011	3	-0.938	-0.941	3
0100	4	-1.250	-1.253	3
0101	5	-1.563	-1.565	2
0110	6	-1.875	-1.880	5
0111	7	-2.188	-2.190	2
1000	8	-2.500	-2.504	4
1001	9	-2.813	-2.816	3
1010	10	-3.125	-3.130	5
1011	11	-3.438	-3.441	3
1100	12	-3.750	-3.755	5
1101	13	-4.063	-4.069	6
1110	14	-4.375	-4.381	6
1111	15	-4.688	-4.693	5

9.5 Viva-Voce Questions & Technical Answers

- **Q1: Why is the measured analog output voltage negative?**
Answer: Because the op-amp is wired as an inverting summing amplifier with negative feedback. The input current enters

virtual ground (0 V) and passes through feedback resistor R_f , creating an output voltage of opposite polarity: $V_{out} = -I_{in} \cdot R_f$. Adding a second unity-gain inverting op-amp restores a positive output.

- **Q2: What is the settling time of a DAC?**

Answer: The elapsed time between the digital input code transition and the moment the analog output voltage settles and remains within $\pm (1 / 2) \text{ LSB}$ of its final steady-state value.
